Chapter 8 Techniques of Integration

8.1 Basic Integration Techniques

$\frac{d}{dx}F(x) = f(x)$

Recall that if $F(x)$ is a differential function of x, and that $F'(x) = f(x)$, then

$$\int f(x)dx = F(x) + C.$$

C: constant.

Recall also that if we let $u = g(x)$, then $du = g'(x)\,dx$

$$\int f(g(x))g'(x)dx = \int f(u)du.$$

TABLE 8.1 Basic integration formulas

1. $\int k\,dx = kx + C$ (any number k)

2. $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \qquad (n \neq -1)$

3. $\int \frac{dx}{x} = \ln|x| + C$

4. $\int e^x\,dx = e^x + C$

5. $\int a^x\,dx = \frac{a^x}{\ln a} + C \qquad (a > 0,\ a \neq 1)$

6. $\int \sin x\,dx = -\cos x + C$

7. $\int \cos x\,dx = \sin x + C$

8. $\int \sec^2 x\,dx = \tan x + C$

8. $\int \csc^2 x\,dx = -\cot x + C$

10. $\int \sec x \tan x\,dx = \sec x + C$

11. $\int \csc x \cot x\,dx = -\csc x + C$

12. $\int \tan x\,dx = \ln|\sec x| + C$

13. $\int \cot x\,dx = \ln|\sin x| + C$

14. $\int \sec x\,dx = \ln|\sec x + \tan x| + C$

15. $\int \csc x\,dx = -\ln|\csc x + \cot x| + C$

16. $\int \sinh x\,dx = \cosh x + C$

17. $\int \cosh x\,dx = \sinh x + C$

18. $\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C$

19. $\int \frac{dx}{a^2 + x^2} = \frac{1}{a}\tan^{-1}\left(\frac{x}{a}\right) + C$

20. $\int \frac{dx}{x\sqrt{x^2 - a^2}} = \frac{1}{a}\sec^{-1}\left|\frac{x}{a}\right| + C$

21. $\int \frac{dx}{\sqrt{a^2 + x^2}} = \sinh^{-1}\left(\frac{x}{a}\right) + C \qquad (a > 0)$

22. $\int \frac{dx}{\sqrt{x^2 - a^2}} = \cosh^{-1}\left(\frac{x}{a}\right) + C \qquad (x > a > 0)$

$\frac{d}{dx}\tan x = \sec^2 x$

$\frac{d}{dx}\sinh x = \cosh x$

$\int \cosh x = \sinh x + C$

23. $\int \frac{1}{a^2 - x^2}\,dx = \frac{1}{a}\tanh^{-1}\left(\frac{x}{a}\right) + C, \quad |x| < |a|,\ x^2 < a^2$

Example 1: Making a Simplifying Substitution

$\int \frac{2x-9}{\sqrt{x^2-9x+1}}dx = 2\sqrt{x^2 - 9x + 1} + C$, by choosing $u = x^2 - 9x + 1$.

$du = (2x-9)\,dx$

$\int \frac{du}{\sqrt{u}} = 2\sqrt{u} + C$

$= 2\sqrt{x^2-9x+1} + C$

Example 2: Completing the Square $\int \frac{dx}{\sqrt{6x - x^2}} = \sin^{-1}\left(\frac{x-3}{3}\right) + C$

$6x - x^2 = -(x^2 - 6x)$

$= -[(x-3)^2 - 9]$

$= 9 - (x-3)^2$

$\int \frac{dx}{\sqrt{9-(x-3)^2}} = \int \frac{du}{\sqrt{a^2-u^2}}$

$a^2 - u^2$

$u = (x-3)$

$du = dx$

$a = 3$

$= \sin^{-1}\left(\frac{u}{a}\right) + C$

$= \sin^{-1}\left(\frac{x-3}{3}\right) + C$

Example 3: Expanding a Power and Using a Trigonometric Identity

$$\int (\sec x + \tan x)^2 dx = 2\tan x + 2\sec x - x + C$$

$$\int \sec^2 x + 2\sec x \tan x + \tan^2 x \, dx$$

$$= \int \sec^2 x \, dx + 2\int \sec x \tan x \, dx + \int \tan^2 x \, dx \qquad (\tan^2 x = \sec^2 x - 1)$$

$$= \tan x + 2\sec x + \int (\sec^2 x - 1) \, dx = \tan x + 2\sec x + \tan x - x + C$$

Example 4: Eliminating a Square Root

$$\int_0^{\frac{\pi}{4}} \sqrt{1 + \cos 4x} \, dx = \int_0^{\frac{\pi}{4}} \sqrt{2\cos^2 2x} \, dx$$

$\cos 2\theta = 2\cos^2\theta - 1$

$1 + \cos 4x = 1 + [2\cos^2 2x - 1]$

$= 2\cos^2 2x$

$$= \int_0^{\frac{\pi}{4}} \sqrt{2}\,|\cos 2x| \, dx$$

(Note that $\cos 2x \geq 0$ when $x \in [0, \frac{\pi}{4}]$)

$$= \int_0^{\frac{\pi}{4}} \sqrt{2}\cos 2x \, dx$$

$$= \left[\frac{\sqrt{2}}{2}\sin 2x\right]_0^{\frac{\pi}{4}} = \frac{\sqrt{2}}{2}\sin\frac{\pi}{2} = \frac{\sqrt{2}}{2} \;//$$

Example 5: Reducing an Improper Fraction

$$\int \frac{4x^2+3}{2x-1}\,dx =$$

$$\int (2x+1) + \frac{4}{2x-1}\,dx$$

$$2x-1 \overline{)\,4x^2+3}$$

$2x+1$

$4x^2-2x$

$2x+3$

$2x-1$

4

$$= x^2 + x + 4\int \frac{1}{2x-1}\,dx$$

$$= x^2+x+4\,\frac{\ln|2x-1|}{2} + C$$

$$= x^2+x+2\ln|2x-1| + C$$

$$\Rightarrow \frac{4x^2+3}{2x-1} = (2x+1) + \frac{4}{2x-1}$$

Example 6: Separating a Fraction

$$\int \frac{1-4x}{x^2+9}\,dx = \int \frac{1}{x^2+9}\,dx - \int \frac{4x}{x^2+9}\,dx$$

u^2+a^2

$u=x,\ a=3$

$u = x^2+9$

$du = 2x\,dx$

$$= \frac{1}{3}\tan^{-1}\left(\frac{x}{3}\right) - \int \frac{2\,du}{u}$$

$$= \frac{1}{3}\tan^{-1}\left(\frac{x}{3}\right) - 2\ln|u|$$

$$= \frac{1}{3}\tan^{-1}\left(\frac{x}{3}\right) - 2\ln|x^2+9| + C$$

$$= \frac{1}{3}\tan^{-1}\left(\frac{x}{3}\right) - 2\ln(x^2+9) + C$$

$(\because\ x^2+9 > 0)$

Extra Examples:

1. $\int \frac{dx}{\sqrt{8-e^{2x}}}$

2. $\int_1^e \frac{dx}{x\sqrt{4+(\ln x)^2}}$

3. $\int_0^{\ln\sqrt{3}} \frac{e^x}{1+e^{2x}}\,dx$

4. $\int_{\sqrt{2}}^{3} \frac{2x^3}{x^2-1}\,dx$

5. $\int_0^{\pi} \sqrt{1-\cos 4x}\,dx$

6. $\int \frac{dx}{1+\sin x}$

$\cos 2\theta = 2\cos^2\theta - 1$

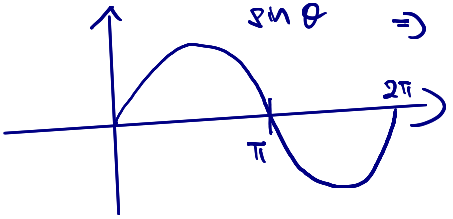
$= 1 - 2\sin^2\theta$

$= \cos^2\theta - \sin^2\theta$

⑤ $\int_0^{\pi} \sqrt{1-\cos 4x}\,dx$

$1-\cos 4x = 1-(1-2\sin^2 2x) = 2\sin^2 2x$

$\int_0^{\pi} \sqrt{2\sin^2 2x}\,dx = \int_0^{\pi} \sqrt{2}\,|\sin 2x|\,dx$



$\sin 2x \geq 0$ when $x \in [0, \frac{\pi}{2}]$

& $\sin 2x \leq 0$ when $x \in [\frac{\pi}{2}, \pi]$

$= \int_0^{\frac{\pi}{2}} \sqrt{2}\sin 2x\,dx + \int_{\frac{\pi}{2}}^{\pi} \sqrt{2}(-\sin 2x)\,dx$

equal

$= 2\int_0^{\frac{\pi}{2}} \sqrt{2}\sin 2x\,dx = 2\sqrt{2}\left[-\frac{\cos 2x}{2}\right]_0^{\frac{\pi}{2}} = \underline{\underline{2\sqrt{2}}}$

(6) $\int \frac{1}{1+\sin x}\,dx = \int \frac{1}{1+\sin x} \times \frac{1-\sin x}{1-\sin x}\,dx$

Multiply the integral with $\frac{1-\sin x}{1-\sin x}$

$= \int \frac{1-\sin x}{1-\sin^2 x}\,dx$

$= \int \frac{1-\sin x}{\cos^2 x}\,dx$

$= \int \frac{1}{\cos^2 x}\,dx - \int \frac{\sin x}{\cos^2 x}\,dx$

$= \int \sec^2 x\,dx - \int \tan x \sec x\,dx$

$= \tan x - \sec x + C$

(i) Find the derivative of $\tanh^{-1} x$, $|x| < 1$.

(ii) Determine $\int \frac{\cos x}{4 - \sin^2 x}\, dx$.

$$\tanh x = \frac{\sinh x}{\cosh x}$$

$$\frac{d}{dx}\tanh x = \frac{(\cosh x)\frac{d}{dx}(\sinh x) - (\sinh x)\frac{d}{dx}(\cosh x)}{(\cosh x)^2} \qquad (\cosh^2 x - \sinh^2 x = 1)$$

$$(1 - \tanh^2 x = \operatorname{sech}^2 x)$$

$$= \frac{\cosh^2 x - \sinh^2 x}{\cosh^2 x} = \frac{1}{\cosh^2 x} = \operatorname{sech}^2 x$$

$$y = \tanh^{-1} x \Rightarrow \tanh y = x$$

$$\Rightarrow \operatorname{sech}^2 y \frac{dy}{dx} = 1 \quad \text{(diff wrt } x\text{)}$$

$$\Rightarrow \frac{dy}{dx} = \frac{1}{\operatorname{sech}^2 y} = \frac{1}{1 - \tanh^2 y} = \frac{1}{1 - x^2}$$

$$\frac{d}{dx}\tanh^{-1}u = \frac{1}{1-u^2}\frac{du}{dx}, \quad |u|<1.$$

integrate wrt x $\Rightarrow$ $\int \frac{1}{1-u^2}\,du = \tanh^{-1}u + C$

$u = \frac{x}{a}$ $\Rightarrow$ $\int \frac{1}{1-\left(\frac{x}{a}\right)^2}\frac{dx}{a} = \tanh^{-1}\frac{x}{a} + C$

$du = \frac{1}{a}dx$

$$\Rightarrow \int \frac{a}{a^2-x^2}\,dx = \tanh^{-1}\left(\frac{x}{a}\right) + C$$

(ii) $\int \frac{\cos x}{4-\sin^2 x}\,dx$ $\Rightarrow$

$$\boxed{\int \frac{1}{a^2-x^2}\,dx = \frac{1}{a}\tanh^{-1}\left(\frac{x}{a}\right) + C}$$

$w = \sin x$

$dw = \cos x\,dx$

$$\int \frac{dw}{4-w^2} = \frac{1}{2}\tanh^{-1}\left(\frac{w}{2}\right) + C = \frac{1}{2}\tanh^{-1}\left(\frac{\sin x}{2}\right) + C$$

$a = 2$

Exercise: $\int \frac{1}{5-4x-x^2}\,dx$, $|x+2|<3$.